

Solution Set

A system of equations places several constraints on the same variables at once. A **solution** is an ordered list of values — one for each variable — that satisfies *every* equation in the system simultaneously, not just one of them. The **solution set** is the collection of all such lists. Depending on the system, this set might contain exactly one solution, no solutions at all, or infinitely many.

Worked example. Consider the system

$$\begin{aligned}x + y &= 10 \\ 2x - y &= 2\end{aligned}$$

A single equation like $x + y = 10$ has infinitely many solutions on its own — $(4, 6)$, $(7, 3)$, $(0, 10)$, and so on. But we need pairs (x, y) that work in *both* equations. Adding the two equations eliminates y : $3x = 12$, so $x = 4$. Substituting back gives $y = 6$. Checking: $4 + 6 = 10$ ✓ and $2(4) - 6 = 2$ ✓. The solution set here is a single ordered pair, written $\{(4, 6)\}$. Geometrically, each equation is a line, and $(4, 6)$ is the one point where the two lines cross.

Problem-solving application. The size of a solution set tells you something important about the system before you even finish solving it. Two lines in a plane either cross once (one solution), never cross because they're parallel (no solution, or the *empty set* $\emptyset$), or lie exactly on top of each other (infinitely many solutions). This matters in practice: if you're modeling a real situation — say, two companies' pricing plans as functions of usage — and you solve the system to find where costs are equal, an empty solution set tells you the plans *never* cost the same, while infinitely many solutions would mean the plans are identical for every amount of usage.

When working with larger systems (three or more variables), the same logic scales up: a solution is now a longer ordered list, such as $(x, y, z) = (1, -2, 5)$, and you verify it the same way — plug it into *every* equation and confirm all are true. Before trusting an answer, always check it against each original equation; a value that satisfies one equation but not another is not part of the solution set, no matter how it was derived.

